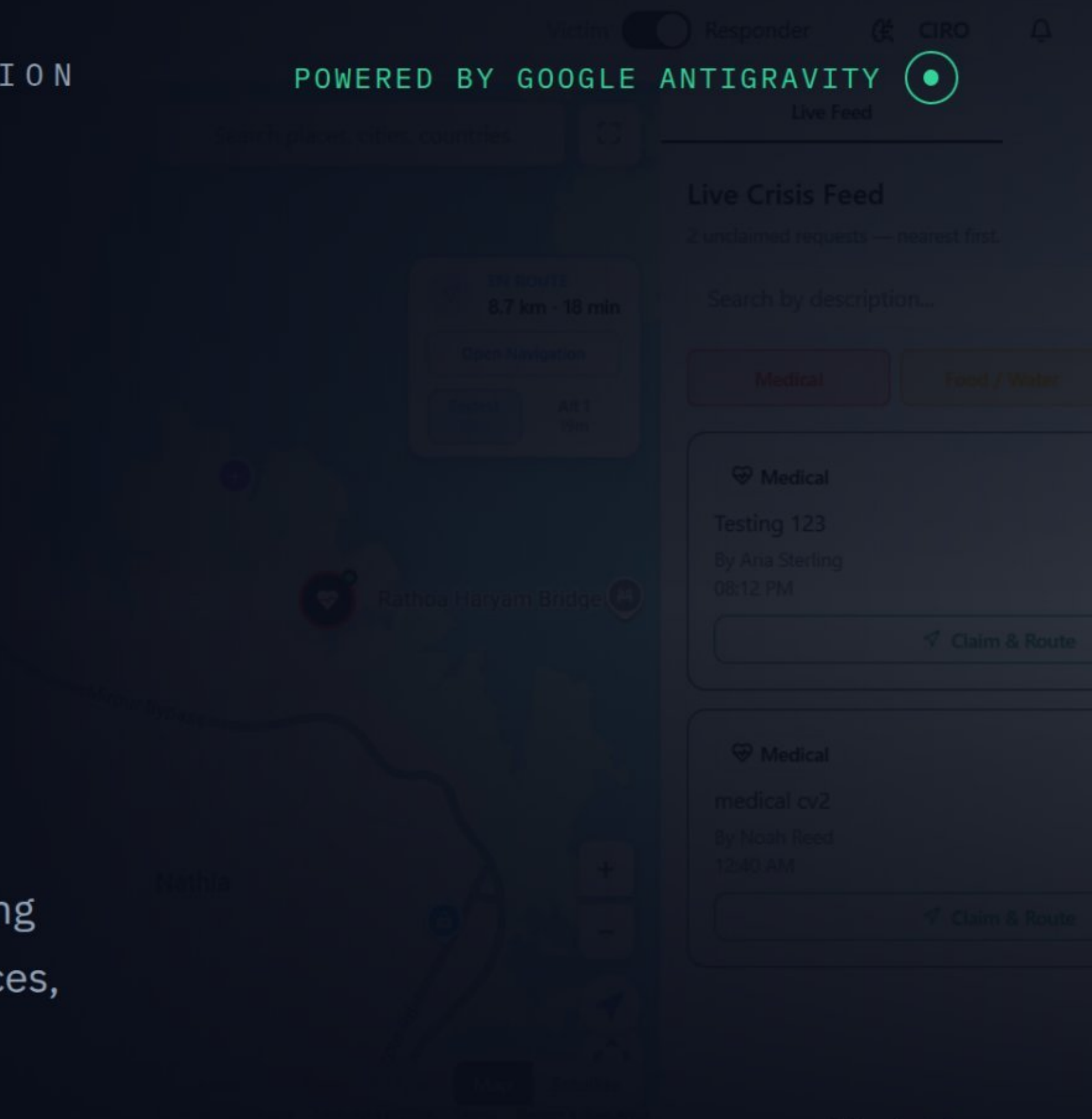




CrisisGrid

Crisis Intelligence & Response Orchestrator

A four-agent Antigravity pipeline that detects emerging emergencies, allocates constrained response resources, and coordinates field action in real time.



When a crisis hits, **coordination breaks** before help arrives.

Signals are scattered across social posts, weather feeds, traffic APIs, IoT sensors, field reports, and emergency calls. Dispatchers triage by hand. Resources are double-allocated. Minutes are lost.

SIGNAL OVERLOAD

Six raw data streams, none
none of them aligned.

MANUAL TRIAGE

Operators classify severity
severity from instinct, not
evidence.

CONSTRAINED RESOURCES

Ambulances and rescue
teams split across competing
incidents with no shared
plan.

One pipeline. **Four agents.** A coordinated answer in seconds.

AGENT 01

Sentinel

SIGNAL FUSION

Ingests six raw streams.
Scores credibility. Flags
contradictions. Outputs
one fused situational
picture.

INIT → NOISE_FILTER →
FUSION_COMPLETE

AGENT 02

Analyst

SEVERITY PREDICTION

Classifies crisis type.
Returns severity 1–5
with confidence.
Quantifies uncertainty
and spread risk.

SEVERITY_CALC →
TEMPORAL_ANALYSIS → HANDOFF

AGENT 03

Strategist

RESOURCE ALLOCATION

Splits units across
simultaneous crises.
Documents trade-offs.
Marks infeasible actions
instead of hiding them.

CONSTRAINT_CHECK →
TRADE_OFF_LOG → ACTION_CHAIN

AGENT 04

Executor

ACTION SIMULATION

Runs simulated tool calls
with retry and fallback.
Computes before/after
state. Issues alerts to five
stakeholder.

TOOL_CALL → FALLBACK_USED →
OUTCOME_COMPUTED

Every reasoning step is **visible**,
visible, time-stamped, and
and exportable.

```
SENTINEL12:04:18NOISE_FILTER · 2 of 5 signals discarded  
SENTINEL12:04:19CONTRADICTION · social vs field officer  
ANALYST12:04:21SEVERITY_CALC · Tier 5 · confidence 0.95  
STRATEGIST12:04:24TRADE_OFF · 3 of 4 ambulances → flood  
EXECUTOR12:04:27TOOL_CALL · dispatch_rescue_unit() ok  
EXECUTOR12:04:31ALERTS_SENT · 5 audiences · COMPLETE
```

38

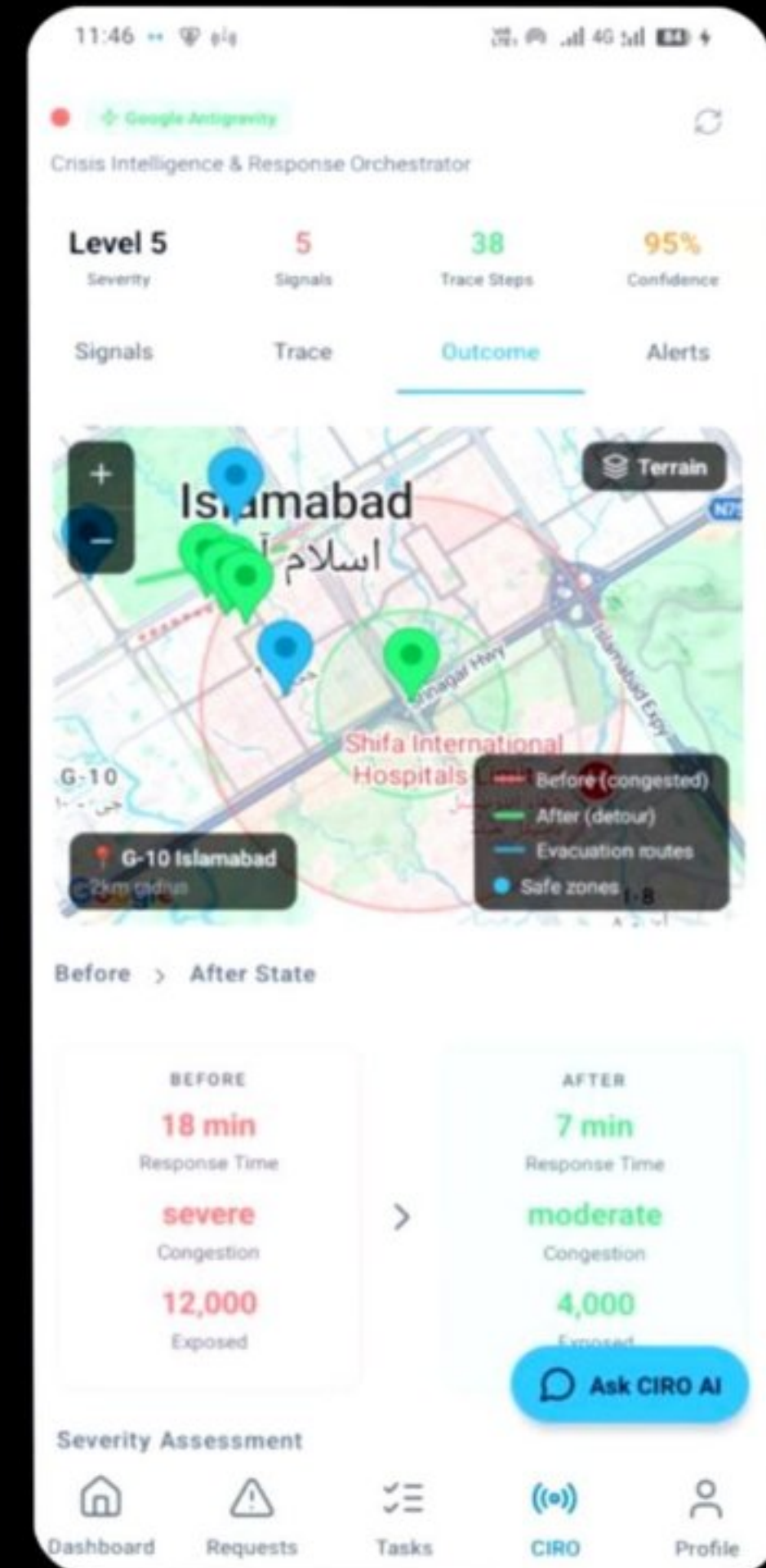
TRACE STEPS

5

SIGNAL SOURCES

95%

CONFIDENCE



We tested CIR0 against the failure modes that **break real systems**.

ID	SCENARIO	SIGNALS	WHAT IT PROVES
01	Urban Flooding G-10 Islamabad	5	Multi-source fusion plus contradiction handling — social posts vs. field officer reporting only a broken water main.
02	Simultaneous Crises Flood + Heatwave	8	Two crises, one resource pool. Strategist must split allocations and log explicit trade-off reasoning.
03	False Alarm Panic vs. verification	4	High severity gets downgraded mid-flight. Executor retracts the public alert and issues corrections to all five audiences.
04	Degraded Mode Traffic API 503	4	Live traffic stream fails. apiClient retries, activates an 18-minute NTRC cache fallback, and continues the pipeline.

One map. Every active crisis. Every volunteer.

- Gaussian heatmap with per-type gradients
- Admin-drawn safe zones, polygon-aware
- Auto-dispatch to nearest on-duty volunteer
- Live volunteer GPS, 5-second cadence

COVERAGE GAP WATCH

every 30s

AUTO-DISPATCH RETRY

QUEUED → CLAIMED

Volunteers receive a task in **seconds**, not minutes.

01

Auto-assignment with ETA

Nearest on-duty responder is claimed atomically. Push notification deep-links straight to the task.

02

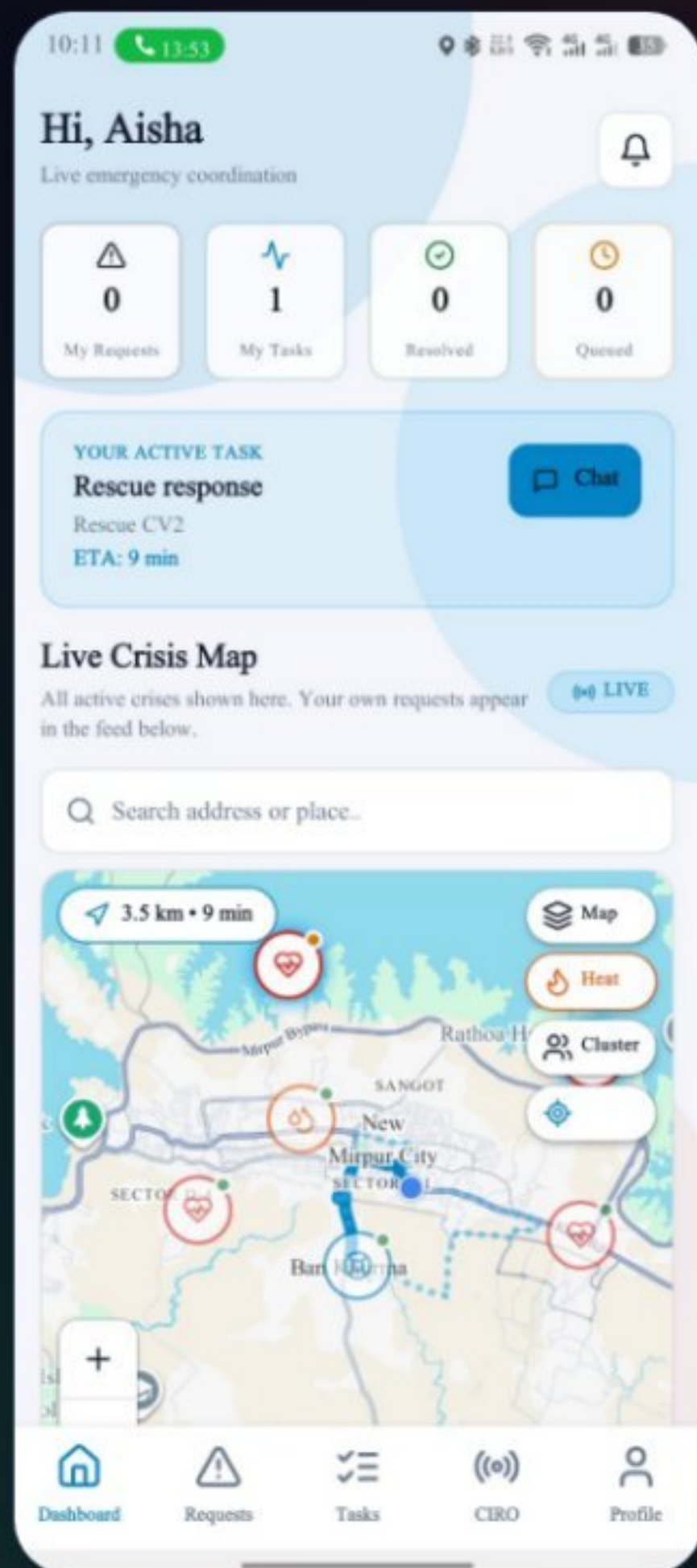
Live crisis map

Heatmap, clustering, and routing built on react-native-maps with Google providers.

03

Two-way chat per request

A chat room opens on claim. Victim, volunteer, and admins see the same thread in real time.



A 25-second pipeline that never **blocks** a request.

REQUEST LIFECYCLE

POST /api/ciro/analyze

- └ returns 202 { jobId } in ms
- └ **setImmediate()** background pipeline
ingest → Sentinel → Analyst → Strategist → Executor

GET /api/ciro/job/:jobId

- └ client polls every 3s → status + result

WS ciro:{sessionId}

- └ live AgentLogEntry stream per step

Jobs garbage-collected after 1 hour · debug log at /api/ciro/api-calls

RESILIENCE

Retry + cached fallback per stream;
failures recorded in the trace, not
swallowed.

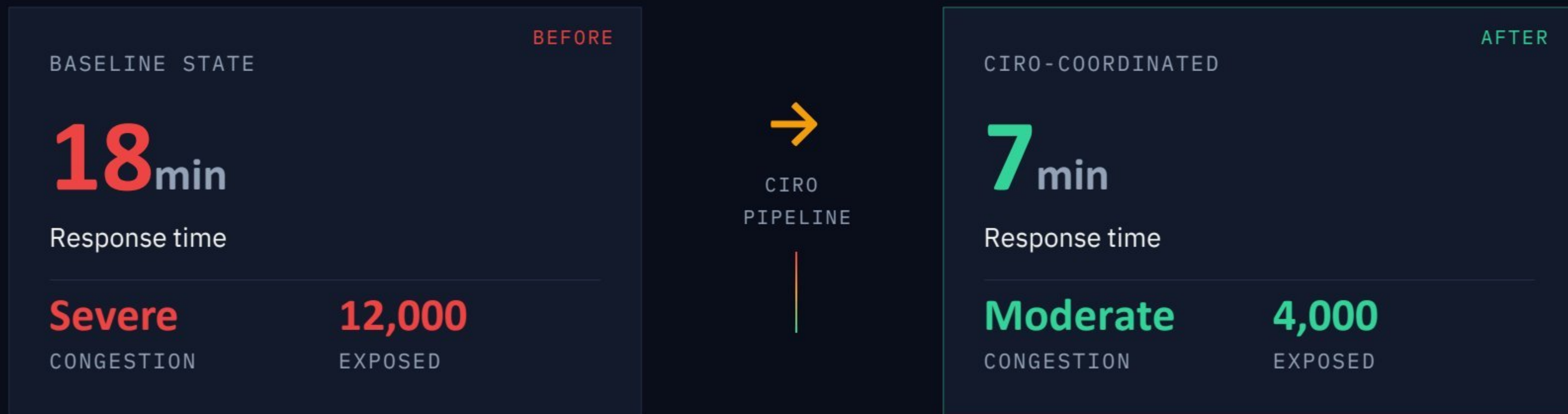
AUTH & TRUST

JWT sessions, TOTP 2FA, role-scoped
routes, full audit log.

STAKEHOLDER REACH

Tailored alerts to Public, Hospital,
Utility, Transport, Media.

The Executor returns a **measurable** change in system state.



CHALLENGE 03

Faster signals. **Better** decisions. Lives saved.

CrisisGrid pairs a multi-source signal grid with a four-agent Antigravity pipeline — built for the minutes that matter.

Stack

React 18 · Expo · Express · Firestore · Socket.IO · OpenAI GPT-4o-mini · Google Maps